CORRECTING SAMPLE SELECTION BIAS WITH CATEGORICAL OUTCOMES

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Abstract. In this paper, I propose a method for correcting sample selection bias when the outcome of interest is categorical, such as occupational choice, health status, or field of study. Classical approaches to sample selection rely on strong parametric distributional assumptions, which may be restrictive in practice. I develop a local representationthat decomposes each joint probability into marginal probabilities and a category-specific association parameter that captures how selection differentially affects each outcome. Under some exclusion restrictions, I establish nonparametric point identification of the latent categorical distribution. Building on this identification result, I introduce a semiparametric multinomial logit model with sample selection, propose a computationally tractable two-step estimator, and derive its asymptotic properties. I illustrate the method by studying the determinants of healthcare utilization in Côte d'Ivoire.

Keywords: Sample selection, categorical data, bivariate logistic, local representation

JEL codes: C14, C26, C35

1. Introduction

Sample selection is a common challenge in applied microeconometrics, arising whenever the outcome of interest is observed only for a non-random selected subset of the population. Ignoring the selection mechanism generally distorts the empirical distribution of the observed outcome, leading to biased estimates and misleading inference. Classical econometric methods developed to address this bias were designed primarily for scalar outcomes, such as continuous wages or binary employment indicators (see the survey by Vella (1998)). However, in many empirical contexts the outcome is categorical and unordered. Examples include employment status (employed, unemployed, inactive), educational attainment (high school, college, postgraduate), migration destinations, health status categories, or multi-option survey responses. In all these cases, selection not only affects the mean outcome but also the probability distribution over the outcome categories. Standard correction methods, developed for scalar or ordered outcomes, cannot be directly applied because the distribution of unordered categories does not admit a natural cumulative representation.

The problem of sample selection bias has been a central concern in econometrics since the seminal work of Heckman (1979). In his classic model, Heckman proposed a two-step correction for continuous outcomes under the assumption that the unobserved errors in the selection and outcome equations are jointly normally distributed. This parametric assumption yields a closed-form expression for the conditional expectation of the outcome given selection—the well-known inverse Mills ratio, which can be included as a control function in the second-stage regression. While elegant and widely applied, this approach is critically dependent on the joint normality assumption; misspecification can lead to inconsistent estimates, as demonstrated by Little and Rubin (1989, 2019) and later by Newey (2009), who showed that even mild deviations from normality can severely bias structural parameters. Extensions to discrete outcomes soon followed. For binary response models, Van de Ven and Van Praag (1981) and Dubin and McFadden (1984); Dubin and Rivers (1989) developed analogues of Heckman's model by assuming bivariate normality of latent indices underlying both selection and the binary outcome. These models remain popular in applied work due to their computational tractability, but they inherit the same vulnerability to distributional misspecification.

Recognizing these limitations, a strand of the literature has pursued semiparametric or nonparametric approaches. Early contributions by Ahn and Powell (1993) and Das et al. (2003) developed estimators that relax distributional assumptions by exploiting instrumental variables or independence conditions. More recently, Han and Vytlacil (2017); Han and Lee (2019) has advanced identification and estimation theory for semiparametric selection models, showing that under mean independence and monotonicity conditions, one can identify treatment effects or structural functions without specifying the joint error distribution. Freedman and Sekhon (2010) further emphasized the fragility of parametric corrections in high-stakes applications, warning that endogeneity corrections are only as credible as their identifying assumptions.

The challenge becomes significantly more complex in settings with unordered categorical outcomes, where the dependent variable takes values in a finite set without natural ordering (e.g., occupation choice, mode of transportation, or political affiliation). Here, the multinomial nature of the outcome introduces additional layers of dependence: selection may affect categories asymmetrically, and the correlation between selection and each category may vary in sign and magnitude. Early attempts to extend Heckman-type models to the multinomial case, such as those by Bourguignon et al. (2007) or Puhani (2000), typically imposed strong parametric structures, often assuming multivariate normality of latent utilities and selection errors, coupled with a specific copula (usually Gaussian) to link them. Similarly, Azzalini et al. (2019) propose using skew-normal or other flexible distributions to relax the symmetry of the normal assumption, offering improved fit in some applications. While such innovations increase robustness to marginal misspecification, they still require specifying the full joint distribution of selection and outcome errors.

A fundamental limitation of these approaches is their treatment of dependence as a global feature of the joint distribution. In contrast, empirical settings often exhibit local or category-specific patterns of selection: for instance, a job training program may attract individuals who are more likely to choose high-skill occupations but less likely to choose low-skill ones, implying heterogeneous dependence across categories. Standard multinomial selection models, by tying all categories to a single latent error structure, struggle to capture such heterogeneity.

This tension between the need for flexible dependence modeling and the desire for tractable, identifiable estimation motivates the development of semiparametric frameworks that decouple marginal behavior from dependence structure. Motivated by these concerns, recent work has shifted toward semiparametric and nonparametric identification strategies. A major advance is the Local Gaussian Representation (LGR) introduced by Chernozhukov et al. (2023) (henceforth CFL). Their approach is built around a simple but powerful insight: any joint distribution of two scalar latent variables can be locally approximated by a Gaussian copula with an implicit, data-dependent correlation parameter. This correlation varies across the support of the marginals and captures the local dependence between the latent outcome and the latent selection variable. Identification of the outcome distribution is achieved by assuming that this local correlation parameter is invariant to a binary instrument. Intuitively, the instrument shifts the probability of selection but does not affect the underlying dependence structure between the latent components of the economic model.

Despite its flexibility, the CFL framework is fundamentally tied to ordered outcomes because distribution regression defines the outcome through cumulative events of the form $\{Y \leq y\}$. For unordered categorical outcomes, such cumulative events are meaningless: there is no coherent ordering of categories such as "public sector job," "self-employment," and "informal wage employment." Attempting to impose an artificial order risks distorting interpretation and generating spurious patterns of dependence. Copula-based extensions encounter the same limitation because they continue to rely on a latent scalar underlying the categorical variable. In short, none of the existing nonparametric approaches can simultaneously (i) accommodate unordered categorical outcomes, (ii) allow flexible dependence between selection and outcomes, and (iii) permit point identification under simple exclusion restrictions.

This paper develops a new framework that resolves these limitations by extending the CFL methodology to unordered categorical outcomes. The starting point is a local representation(LR), which provides a general decomposition for any joint probability involving outcome categories and selection. For a given category c, the joint probability

$$\mathbb{P}(Y = c, S = 1)$$

can be decomposed as

$$\mathbb{P}(Y = c, S = 1) = p_c q + \omega_c \sqrt{p_c(1 - p_c)}, \sqrt{q(1 - q)}$$

where (i) $p_c = \mathbb{P}(Y = c)$ is the outcome marginal, (ii) $q = \mathbb{P}(S = 1)$ is the selection rate, and (iii) $\omega_c \in [-1, 1]$ is a local association parameter measuring the dependence between the event $\{Y = c\}$ and selection. This decomposition follows from a simple covariance identity for indicator functions:

$$\mathrm{Cov}(1_{Y=c}, 1_{S=1}) = \omega_c \sqrt{p_c(1 - p_c)} \sqrt{q(1 - q)},$$

allowing the joint probability to be expressed as the independence benchmark plus an interpretable adjustment capturing the extent of local dependence. Unlike cumulative-based representations, this framework applies directly to probabilities and thus avoids any need for an ordering of the outcome categories.

A key advantage of the LR is that it naturally extends to conditional settings. For any vector of covariates (X), the decomposition

$$\mathbb{P}(Y = c, S = 1 \mid X) = p_c(X)q(X) + \omega_c(X) \sqrt{p_c(X)(1 - p_c(X))} \sqrt{q(X)(1 - q(X))}$$

holds identically, where each component, marginals and dependence, may vary freely with X. This conditional representation is especially valuable in econometrics because most identification arguments, exclusion restrictions, and estimation procedures operate conditional on covariates.

Building on this conditional representation, the paper establishes nonparametric point identification of the latent unordered categorical distribution under exclusion restrictions analogous to those in CFL. These restrictions are relatively weak: the association parameter is allowed to differ across categories and covariate values, enabling flexible patterns of category-specific selection. I require that the instrument shifts selection but not the category-specific local dependence between outcome and selection, conditional on observables. This formulation allows the LR to capture rich patterns of heterogeneous dependence across categories and covariate strata, something parametric models are rarely able to accommodate. To operationalize the framework, the paper introduces a semiparametric multinomial logistic regression model with sample selection, constructed in the spirit of the distribution

regression approach but adapted for unordered categorical outcomes. A two-step estimator is developed, and its asymptotic normality is established.

Overall, the paper makes three main contributions. First, it proposes the local representation, a new tool for modeling joint probabilities involving unordered outcomes. Second, it provides nonparametric identification results under exclusion restrictions, extending distribution regression techniques to categorical contexts. Third, it offers a practical semiparametric estimation method that is flexible, interpretable, and computationally efficient. An empirical application to healthcare utilization in Côte d'Ivoire demonstrates the method's usefulness: wealth, education, gender, and urban residence significantly increase the probability of choosing formal healthcare providers over informal ones.

The remainder of the paper is structured as follows. Section 2 develops the identification analysis using the local representation. Section 3 states the main identification results. Section 4 introduces the semiparametric selection model and inference procedure. Section 5 presents the empirical application. Section 6 concludes. Proofs appear in the Appendix.

2. LOCAL REPRESENTATIONOF A JOINT PROBABILITY

The identification approach I propose extends the framework of CFL by introducing a new local representationtheorem that remains valid even when the outcome variable is categorical or unordered. This extension is crucial because most existing identification results based on local representations, including the Local Gaussian Representation (LGR) of Anjos and Kolev (2005), Chernozhukov et al. (2023), and Chernozhukov et al. (2024) implicitly rely on an ordered outcome space. Their constructions use events of the form $\{Y \leq y\}$, which are not meaningful when Y is unordered or nominal.

The contribution here is to provide a local representationthat accommodates both continuous and discrete outcomes, including unordered categorical outcomes commonly encountered in empirical work. Beyond its relevance for the identification problem studied in this paper, the representation theorem is conceptually general and may prove useful in a wide range of econometric applications where modeling local dependence is advantageous.

The key insight is that for any pair of events defined by two random variables, their joint probability can always be decomposed into (i) a component implied by marginal probabilities and (ii) a single, interpretable local dependence parameter. To formalize this, define the

function

$$\Lambda_2(u, v, \omega) = uv + \omega\sqrt{u(1-u)}\sqrt{v(1-v)}, \qquad (u, v, \omega) \in [0,1]^2 \times [-1, 1].$$

Lemma 1 (Local Representation)**.** *Let Y and S be random variables supported on measurable spaces $(\mathcal{Y}, \sigma_Y)$ and $(\mathcal{S}, \sigma_S)$, respectively. For any measurable sets $A \times B \in \sigma_Y \times \sigma_S$ with marginal probabilities*

$$u := \mathbb{P}(Y \in A) \in (0,1), \qquad v := \mathbb{P}(S \in B) \in (0,1),$$

there exists a unique parameter $\omega_{Y,S}(A, B) \in [-1, 1]$ such that

$$\mathbb{P}(Y \in A,\, S \in B) = \Lambda_2\big(u,\, v,\, \omega_{Y,S}(A, B)\big).$$

Moreover,

$$\omega_{Y,S}(A, B) = \mathrm{Corr}\big(\mathbf{1}_{\{Y \in A\}},\, \mathbf{1}_{\{S \in B\}}\big)\,,$$

so that positive (negative) values indicate local positive (negative) dependence, and $\omega = 0$ corresponds to local independence.

Lemma 1 establishes a pointwise decomposition of joint probabilities: for any pair of events (A, B), the quantity $\mathbb{P}(Y \in A, S \in B)$ can be written exactly using only the two marginal probabilities and a single, well-defined local association parameter. This decomposition cleanly separates the independence part, uv, from the dependence adjustment, $\omega\sqrt{u(1-u)}\sqrt{v(1-v)}$.

This result generalizes Theorem 1 in Anjos and Kolev (2005), which is restricted to scalar Y and S and to ordered events of the form $\{Y \leq y\}$ and $\{S \leq s\}$. In contrast, Lemma 1 holds for arbitrary events A and B, regardless of whether Y and S are continuous, discrete, unordered, or vector-valued. The theorem states only that for each pair (A, B), there exists a unique ω matching the observed joint probability; it does not impose any global parametric restrictions. Hence, the local parameter provides a flexible and interpretable measure of how dependence between Y and S moves the joint probability above or below the benchmark of independence.

Finally, this local representationextends naturally to conditional settings. Allowing both marginal probabilities and the local dependence parameter to vary with observed covariates

X yields

$$\mathbb{P}(Y \in A,\, S \in B \mid X) = \Lambda_2\Big(\mathbb{P}(Y \in A \mid X),\, \mathbb{P}(S \in B \mid X),\, \omega_{Y,S}(A, B \mid X)\Big).$$

This property makes the framework particularly valuable for econometric models involving sample selection, treatment effects, or endogenous discrete choices, where identification and estimation typically proceed conditional on covariates. By providing a smooth, interpretable, and universally valid decomposition of joint probabilities, the local representationoffers a powerful foundation for semiparametric identification and estimation in models with discrete, continuous, or mixed-type outcomes.

3. Model and identification

In this section, I introduce the model and the necessary assumptions to identify the distribution of the latent categorical variable. Suppose the latent outcome Y^* takes values in a finite set of q mutually exclusive categories,

$$Y^* \in \{c_1, c_2, \ldots, c_q\}.$$

These categories are unordered (e.g., occupation, marital status, political affiliation). I adopt the usual multinomial parametrization and take the last category c_q as the baseline (without loss of generality). This choice allows us to work with log odds relative to the baseline. Let the marginal distribution of Y^* be $\pi(\pi_1, \ldots, \pi_q) \in \mathcal{S}^{q-1}$, where $\mathcal{S}^{q-1}$ denotes the $(q-1)$-simplex. The log-odds map $\ell : \mathcal{S}^{q-1} \to \mathbb{R}^{q-1}$ and its inverse make the connection between probabilities and real-valued indices: $\ell(\pi) = \big(\log(\pi_1/\pi_q), \ldots, \log(\pi_{q-1}/\pi_q)\big)$, and for any $y \in \mathbb{R}^{q-1}$, $\ell^{-1}(y) = \frac{1}{1+\sum_{i=1}^{q-1} e^{y_i}}\big(e^{y_1}, \ldots, e^{y_{q-1}}, 1\big)$. Equivalently, writing $\mu_k = \log(\pi_k/\pi_q)$ for $k = 1, \ldots, q-1$ (and $\mu_q \equiv 0$ by normalization), each category probability admits the familiar softmax form:

$$\pi_k = \mathbb{P}(Y^* = c_k) = \frac{e^{\mu_k}}{\sum_{j=1}^{q} e^{\mu_j}}, \qquad k = 1, \ldots, q.$$

I now introduce the sample selection mechanism. Let (Y^*, U_z) denote the latent outcome and latent selection variables, $U_z \in \mathbb{R}^p$ may be multivariate, allowing a flexible selection process. For binary instrument $Z \in \{0, 1\}$ define the potential selection indicators

$$S_z = 1_{\{U_z \geq 0\}}, \quad S = S_1 Z + S_0(1 - Z).$$

This formulation generalizes the standard threshold-crossing selection model by accommodating multivariate latent selection determinants U_z. Now I observe

$$Y = Y^* \quad \text{only if } S = 1.$$

Identification will rely on the common set of IV conditions.

Assumption 1. *I make the following assumptions :*

- *Independence* $(Y^*, U_z) \perp Z$.
- *Non-degeneracy:* $0 < \mathbb{P}(S = 1) < 1$ *and* $0 < \mathbb{P}(Z = 1 \mid S = 1) < 1$.
- *Relevance:* $\mathbb{P}(S = 1 \mid Z = 0) < \mathbb{P}(S = 1 \mid Z = 1) < 1$

The identification strategy relies on three key assumptions. Independence is the standard exclusion restriction in the instrumental variables (IV) literature. It requires that the instrument Z is as good as randomly assigned with respect to both the latent outcome Y^* and the latent selection variable U_z. In other words, while Z may influence whether an individual is observed through the selection mechanism, it must not directly affect either the distribution of potential outcomes Y^* or the unobserved determinants of selection. This ensures that the identifying variation arises solely from changes in the selection probability induced by the instrument. The non-degeneracy condition rules out trivial cases. If $\mathbb{P}(S = 1) = 0$ or $\mathbb{P}(S = 1) = 1$, selection is deterministic, leaving no scope to study how outcomes depend on selection. Likewise, if $\mathbb{P}(Z = 1 \mid S = 1)$ equals 0 or 1, then within the selected sample the instrument takes only a single value, providing no useful variation. By guaranteeing that both selection and the instrument vary in the population, non-degeneracy makes the problem statistically informative. Finally, relevance plays the role of a first-stage condition in IV analysis. It requires that the instrument shifts the probability of selection in a non-trivial way. Formally, the strict inequality $\mathbb{P}(S = 1 \mid Z = 0) < \mathbb{P}(S = 1 \mid Z = 1) < 1$ ensures that the instrument is neither weak nor irrelevant. Without such variation, it would be impossible to disentangle the distribution of outcomes from the selection process.

Let $\nu_z = \mathbb{P}(S = 1 \mid Z = z)$ denote the selection probability, and let the logit of the category probability for $k = 1, \ldots, q-1$ be $\pi_k = \frac{e^{\mu_k}}{\sum_{j=1}^{q} e^{\mu_j}}$. By Lemma 1 (the local representation) and Assumption 1, the joint probability of belonging to category c_k and being selected when $Z = z$ can be written, for each non-baseline category k, as

$$\mathbb{P}(Y = c_k,\, S = 1 \mid Z = z) = \mathbb{P}(Y^* = c_k,\, U_z \geq 0) = \Lambda_2(\pi_k,\, \nu_z,\, \omega_{Y^*,U_z}(c_k, 0))\,.$$

Define the category and instrument–specific local association

$$\omega_{k,z} \equiv \omega_{Y^*,U_z}(c_k, 0).$$

Intuitively, $\omega_{k,z}$ measures how selection (conditional on $Z = z$) shifts the odds of category c_k: a positive value means that selection increases the relative probability of c_k, whereas a negative value means that selection reduces it.

Observed components. For each instrument value $z \in \{0, 1\}$, the data reveal: the joint probabilities $\mathbb{P}(Y = c_k,\ S = 1 \mid Z = z)$ for each $k = 1, \ldots, q-1$, and the marginal selection probability $\mathbb{P}(S = 1 \mid Z = z)$. Thus, the data provide a total of $2 \times (q-1) + 1 = 2q - 1$ observable scalar quantities.

Unknown parameters. Before imposing any further restrictions, the model includes: the $q-1$ outcome logits $\mu = (\mu_1, \ldots, \mu_{q-1})$, and the category– and instrument–specific dependence parameters $\omega_{k,z}$ for each $k = 1, \ldots, q-1$ and $z \in \{0, 1\}$, totaling $2(q-1)$ unknown association parameters. Hence, the model contains $3(q-1)$ free parameters.

Identification. The data deliver $2(q-1)$ equations involving the joint probabilities (one for each non-baseline category and instrument value). In contrast, the parameter space consists of $3(q-1)$ unknowns. Thus, without additional structure, the system is underdetermined, and the parameter vector $(\mu, \omega_{k,z})$ is only *partially identified*. In particular, the joint distribution of (Y^*, S) given Z is insufficient to recover both the category logits and the full set of local selection–outcome association parameters. To obtain point identification, I impose an exclusion (sorting) restriction that parallels the strategy in CFL.

Assumption 2. *(Selection–Sorting Exclusion) For each category k, the local association does not depend on the instrument:*

$$\omega_{k,0} = \omega_{k,1} = \omega_k, \qquad k = 1, \ldots, q-1.$$

This assumption requires that the way selection sorts individuals across categories is invariant to the instrument, the instrument affects only the baseline odds of selection (ν_z), not the category-specific dependence between outcome and selection. When Z has more than two values, the assumption needs to hold for at least two values, holding for more provides

overidentifying restrictions and thus empirical tests. The assumption is similar to the exclusion restriction used in CFL, thus extending their identification logic. It remains more flexible than just assuming a constant (which would be making a parametric assumption) since the local parameter varies with the category. Under Assumption 2, the unknowns reduce to the $2 \times (q-1)$, vector $(\mu_1, \ldots, \mu_{q-1}, \omega_1, \ldots, \omega_{q-1})$. These parameters are determined by the $2 \times (q-1)$ equations (one for each z and k):

$$\mathbb{P}(Y = c_k, S = 1 \mid Z = z) = \Lambda_2\big(\pi_k, \nu_z, \omega_k\big), \qquad z \in \{0, 1\},\ k = 1, \ldots, q-1. \tag{1}$$

This system can be shown to have a unique solution. Theorem 1 formalizes point identification.

Theorem 1. *Suppose Assumptions 1 and 2 hold. Further assume the observed joint distribution of (Y, S, Z) satisfies $|\omega_k| < 1$ for all k (interiority condition). Then the parameter vector (μ, ω) is point-identified as the unique interior solution to the nonlinear system (1). Once the parameters μ_k are identified, the full outcome distribution π can be recovered in closed form: $\pi = \frac{1}{1+\sum_{j=1}^{q-1}\exp(\mu_j)}\Big(\exp(\mu_1), \exp(\mu_2), \ldots, \exp(\mu_{q-1}), 1\Big)$.*

Under interiority conditions, the mapping $(\mu_k, \omega_k) \mapsto \Lambda_2(\pi_k(\mu), \nu_z, \omega_k)$ is smooth and strictly monotonic in each argument. Critically, variation in ν_z, the instrument, induced shifts in selection log-odds, provides the identifying leverage necessary to simultaneously recover both the marginal category logits $\pi_k(\mu)$ (and hence μ_k) and the category-specific association parameter ω_k. In essence, the local representation (LR) reduces the sample selection problem with unordered categorical outcomes to a tractable system of logistic-type equations: the marginal logits μ and the selection-sorting parameters ω are jointly identified from instrument driven variation in selection probabilities, provided the dependence structure (i.e., ω_k) remains stable across instrument values. In the next section, I develop a multinomial logistic regression model that is based on the identification result.

4. Semiparametric multinomial logistic regression model with sample selection

In this section, I develop a flexible semiparametric framework to correct for sample selection bias in settings with unordered categorical outcomes. The proposed approach combines the interpretability of parametric models with the robustness of nonparametric dependence

structures, leveraging the local representation introduced in Lemma 1. The estimation procedure is computationally straightforward, relying on a two-step algorithm that separates the modeling of selection from the joint modeling of outcomes and selection-induced sorting. I also establish the asymptotic properties of the resulting estimators, showing consistency and asymptotic normality under standard regularity conditions.

4.1. **Estimation Procedure.** Let $Y^* \in \{c_1, \ldots, c_q\}$ denote a categorical outcome variable with $q \geq 2$ unordered categories, and let $S \in \{0, 1\}$ be a binary sample selection indicator: $S = 1$ if the individual is observed, and $S = 0$ otherwise. The analyst observes only the sub-sample for which $S = 1$. Let $W = (X, Z)$ denote a vector of covariates, where $X \in \mathbb{R}^{d_x}$ contains exogenous control variables, and $Z \in \{0, 1\}$ is a binary instrumental variable that affects selection but is excluded from the outcome equation, a key identifying assumption.

The goal is to recover the marginal distribution of Y^* in the full population (i.e., unconditional on selection) while accounting for potential nonignorable selection. To this end, I specify a coherent joint model for (Y, S) that satisfies three conditions: (i) it parameterizes the marginal distribution of Y via a multinomial logit link; (ii) it models the selection probability $\mathbb{P}(S = 1 \mid W)$ using a standard logit; and (iii) it captures the dependence between Y and S through the local association parameter $\omega_k(X)$ implied by Lemma 1. This construction generalizes classical selection models (e.g., Heckman's probit) to the multinomial setting while preserving tractability.

For each non-baseline category $k = 1, \ldots, q-1$, define the multinomial logit index as

$$\mu_k(X) = \log\left(\frac{\mathbb{P}(Y = c_k \mid X)}{\mathbb{P}(Y = c_q \mid X)}\right) = X'\beta_k,$$

with the baseline category c_q normalized so that $\mu_q(X) = 0$. The implied conditional choice probabilities are

$$\pi_k(X) = \mathbb{P}(Y = c_k \mid X) = \frac{\exp(X'\beta_k)}{1 + \sum_{j=1}^{q-1} \exp(X'\beta_j)}, \quad k = 1, \ldots, q-1,$$

where $\pi_q(X) = 1 - \sum_{k=1}^{q-1} \pi_k(X)$. This logit specification is standard, well-understood, and ensures that the outcome margins are smoothly parameterized by $\beta = (\beta_1, \ldots, \beta_{q-1})$. Turning to selection, I assume a logistic model:

$$\nu(W) = \mathbb{P}(S = 1 \mid W) = \Lambda(W'\delta), \quad \text{where} \quad \Lambda(t) = \frac{e^t}{1 + e^t}.$$

The instrument Z enters only through W, and its exclusion from the outcome equation is formalized in the identification assumptions.

Critically, the joint probability $\mathbb{P}(Y = c_k, S = 1 \mid W)$ is not assumed to factorize as $\pi_k(X)\nu(W)$ (which would imply conditional independence). Instead, I invoke the local representation of Lemma 1, which guarantees the existence of a unique local association parameter $\omega_k(X) \in [-1, 1]$ such that

$$\mathbb{P}(Y = c_k, S = 1 \mid W) = \Lambda_2\big(\pi_k(X), \nu(W), \omega_k(X)\big),$$

where

$$\Lambda_2(u, v, \omega) = uv + \omega\sqrt{u(1-u)v(1-v)}.$$

To ensure parsimony and smoothness, I parameterize the local association as

$$\omega_k(X) = \tanh(X'\gamma_k), \quad \tanh(u) = \frac{e^u - e^{-u}}{e^u + e^{-u}} \in [-1, 1].$$

with $\tanh(\cdot)$ the hyperbolic tangent function, which maps $\mathbb{R}$ onto $(-1, 1)$. This guarantees that $\omega_k(X)$ remains within the admissible correlation bounds for all X. The vector γ_k captures how covariates modulate the degree of selection-induced sorting into category c_k: a positive $\omega_k(X)$ indicates that, conditional on X, selected individuals are more likely to belong to c_k than the marginal probability $\pi_k(X)$ would suggest; a negative value implies under-representation. Putting the pieces together, the structural model for the observable joint probability is

$$\mathbb{P}(Y = c_k, S = 1 \mid W) = \Lambda_2\left(\frac{\exp(X'\beta_k)}{1 + \sum_{j=1}^{q-1}\exp(X'\beta_j)},\ \Lambda(W'\delta),\ \tanh(X'\gamma_k)\right).$$

Given an i.i.d. sample $\{(Y_i, S_i, X_i, Z_i)\}_{i=1}^n$, the observed-data log-likelihood is

$$\ell_n(\beta, \gamma; \delta) = \sum_{i=1}^{n}\sum_{k=1}^{q} S_i \mathbf{1}\{Y_i = c_k\} \log \Lambda_2\left(\pi_{ik}, \nu_i, \omega_{ik}\right),$$

where $\pi_{ik} = \pi_k(X_i)$, $\nu_i = \Lambda(W_i'\delta)$, and $\omega_{ik} = \tanh(X_i'\gamma_k)$.

I propose the following two-step estimator, outlined in Algorithm 1, which extends Heckman's classical correction to the multinomial-logit context while preserving computational simplicity.

Algorithm 1. *(Two-Step Estimation Procedure)*

- ***Step 1: Estimate the selection equation.*** *Estimate δ by maximizing the standard binary logit likelihood:*

$$\hat{\delta} = \arg\max_{p \in \mathbb{R}^{d_w}} \sum_{i=1}^{n} \Big[S_i \log \Lambda(W_i' p) + (1 - S_i) \log(1 - \Lambda(W_i' p)) \Big].$$

  This yields consistent and asymptotically normal estimates $\hat{\nu}_i = \Lambda(W_i' \hat{\delta})$ of the selection probabilities under standard conditions.

- ***Step 2: Estimate outcome and sorting parameters.*** *For each category $k = 1, \ldots, q-1$, define $\theta_k = (\beta_k, \gamma_k)$, and let $\theta = (\theta_1, \ldots, \theta_{q-1})$. The second-stage estimator solves*

$$\hat{\theta} = \arg\max_{t} \sum_{i=1}^{n} \sum_{k=1}^{q} S_i \mathbf{1}\{Y_i = c_k\} \log \Lambda_2 \left(\pi_k(X_i; t), \hat{\nu}_i, \tanh(X_i' g_k) \right),$$

  where $\pi_k(X_i; t)$ is computed using the candidate β-components of t, and g_k denotes the γ_k-components. The optimization is smooth and can be implemented using standard quasi-Newton methods (e.g., BFGS). Because the likelihood separates across categories only through shared covariates (not parameters), joint estimation is recommended to account for cross-category dependence in the information matrix.

This two-step approach is attractive for several reasons. First, it avoids the computational burden of full maximum likelihood while retaining consistency under correct specification. Second, the use of a logit link in both the selection and outcome equations ensures compatibility with the Λ_2 representation, which is derived from binary correlations. Third, the tanh parameterization of $\omega_k(X)$ enforces the natural bounds of the correlation coefficient without requiring constrained optimization. Under standard regularity conditions, such as compact parameter spaces, identification (Theorem 1), and smoothness of the log-likelihood, the two-step estimator $(\hat{\beta}, \hat{\gamma})$ is consistent and asymptotically normal. The asymptotic variance can be consistently estimated using the usual sandwich formula, with the first-step estimation of δ contributing an additional term that vanishes if one employs a joint estimation approach or

accounts for the first-step variability via the delta method. In practice, bootstrap inference provides a robust alternative.

4.2. **Inference.** In this section, I derive the asymptotics of our proposed estimator and also how inference can be performed using a multiplier bootstrap procedure.

4.2.1. *Asymptotic Normality of the Two-Step Estimator.* The estimator $\hat{\theta}$ is a two-step M-estimator, defined as the maximizer of the sample criterion

$$\hat{\theta} = \arg \max_{t=\{(\beta_k,\gamma_k)\}_{k=1}^{q-1}} \sum_{i=1}^{n} \ell_i(t, \hat{\delta}),$$

where the second-stage log-likelihood contribution for observation i is

$$\ell_i(t, \hat{\delta}) = \sum_{k=1}^{q} S_i \mathbf{1}\{Y_i = c_k\} \log \Lambda_2\Big(\pi_k(X_i;\beta),\, \nu_i,\, \omega_{ik}\Big),$$

with

$$\pi_k(X_i;\beta) = \frac{\exp(X_i'\beta_k)}{1 + \sum_{j=1}^{q-1} \exp(X_i'\beta_j)}, \quad \nu_i = \Lambda(W_i'\hat{\delta}), \quad \omega_{ik} = \tanh(X_i'\gamma_k),$$

Because ν_i depends on the first-step estimator $\hat{\delta}$, this constitutes a two-step M-estimator with a generated regressor. The asymptotic distribution accounts for the sampling variability in both steps. Applying the general framework of Newey and McFadden (1994), we obtain the following result.

Theorem 2 (Asymptotic Normality of the Two-Step Estimator)**.** *Assume the following regularity conditions hold:*

- *The data $\{(Y_i, S_i, X_i, Z_i)\}_{i=1}^n$ are i.i.d. and generated by the structural model with true parameters δ_0 and $\theta_0 = (\beta_{10}, \gamma_{10}, \ldots, \beta_{(q-1)0}, \gamma_{(q-1)0})$, where $\beta_{q0} = 0$.*
- *The parameter spaces for δ and θ are compact, and (δ_0, θ_0) lie in the interior.*
- *The selection equation is correctly specified as a logit, and the Fisher information for δ_0 is positive definite:*

$$I_1 = \mathbb{E}[\Lambda(W'\delta_0)(1 - \Lambda(W'\delta_0))\, WW'] \succ 0.$$

- *The map $(\theta, \delta) \mapsto \Lambda_2(\pi_k(X;\beta), \Lambda(W'\delta), \tanh(X'\gamma_k))$ is twice continuously differentiable in a neighborhood of (θ_0, δ_0), with $\pi_k(X;\beta) \in (0,1)$ and $\Lambda(W'\delta) \in (0,1)$ almost surely. The negative expected Hessian with respect to θ is positive definite:*

$$H_2 = -\mathbb{E}\big[\nabla_{\theta\theta}^{\top} \ell_i(\theta_0, \delta_0)\big] \succ 0.$$

- *The cross-derivative matrix exists and is finite:*

$$G = -\mathbb{E}\big[\nabla_{\theta\delta}^{\top}\ell_i(\theta_0, \delta_0)\big] .$$

- *Standard moment and domination conditions hold to justify differentiation under the expectation.*

Then $\hat{\theta}$ is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V_\theta),$$

where the asymptotic variance is

$$V_\theta = H_2^{-1}\left(\Omega + GI_1^{-1}G^{\top}\right)H_2^{-1},$$

with

$$\Omega = \mathbb{E}\big[\psi_{2,i}\psi_{2,i}^{\top}\big] , \quad \psi_{2,i} = \nabla_\theta \ell_i(\theta_0, \delta_0), \quad \psi_{1,i} = (S_i - \Lambda(W_i'\delta_0))W_i.$$

The key innovation lies in the explicit dependence of ℓ_i on Λ_2. Using the chain rule, the score $\psi_{2,i}$ involves derivatives of $\log \Lambda_2$ with respect to $u = \pi_k$, $v = \nu_i$, and $\omega = \omega_{ik}$, multiplied by derivatives of those quantities with respect to θ. Because Λ_2 is smooth and strictly positive on the interior of its domain (guaranteed by the interiority assumption $|\omega_k| < 1$ and non-degenerate marginals), these derivatives exist and are bounded under standard moment conditions. The term $GI_1^{-1}G^{\top}$ captures the asymptotic impact of estimating δ in the first step. Ignoring it would yield inconsistent standard errors, as the variability from $\hat{\delta}$ propagates into $\hat{\theta}$ through the argument $v = \Lambda(W_i'\hat{\delta})$ in Λ_2. While V_θ can be estimated analytically by plugging in sample analogs of H_2, G, and I_1, this requires computing second derivatives of Λ_2, which, though tractable, are algebraically involved. A more robust approach is the multiplier bootstrap, which sidesteps explicit variance estimation.

4.2.2. *Multiplier Bootstrap Inference.* We now describe a multiplier bootstrap that consistently approximates the sampling distribution of $\hat{\theta}$, fully respecting the two-step structure and the form of Λ_2. Let $\hat{\theta}$ be the second-step estimator obtained using $\hat{\nu}_i = \Lambda(W_i'\hat{\delta})$ from the first-step logit. Define the individual log-likelihood contribution as

$$\ell_i(\theta) = \sum_{k=1}^{q} S_i \mathbf{1}\{Y_i = c_k\} \log \Lambda_2\big(\pi_k(X_i; \beta),\, \hat{\nu}_i,\, \tanh(X_i'\gamma_k)\big).$$

The bootstrap proceeds as follows:

Algorithm 2. *(Multiplier Bootstrap)*

(1) Compute the score vector for each observation at $\hat{\theta}$:

$$s_i(\hat{\theta}) = \nabla_\theta \ell_i(\hat{\theta}),$$

(2) Estimate the observed information matrix:

$$\hat{\mathcal{I}}_n = -\frac{1}{n}\sum_{i=1}^{n} \nabla^2_{\theta\theta}\ell_i(\hat{\theta}).$$

For numerical stability, regularize if needed:

$$\hat{\mathcal{I}}_n^{\text{reg}} = \hat{\mathcal{I}}_n + \lambda I_{d_\theta}, \quad \lambda = 10^{-6}\cdot\lambda_{\max}(\hat{\mathcal{I}}_n), \text{ and set } \widehat{\Sigma} = (\hat{\mathcal{I}}_n^{\text{reg}})^{-1}.$$

(3) For $b = 1, \ldots, B$:

(a) Draw i.i.d. multipliers $G_i^{(b)} \sim \mathcal{N}(0,1)$.

(b) Compute the perturbed score average:

$$\bar{s}^{(b)} = \frac{1}{n}\sum_{i=1}^{n} G_i^{(b)} s_i(\hat{\theta}).$$

(c) Form the bootstrap draw:

$$\theta^{(b)} = \hat{\theta} + \widehat{\Sigma}\,\bar{s}^{(b)}.$$

(4) Construct percentile confidence intervals from $\{\theta^{(b)}\}_{b=1}^{B}$.

This procedure is valid because the empirical score $\frac{1}{\sqrt{n}}\sum_i s_i(\hat{\theta})$ consistently estimates the influence function of $\hat{\theta}$, including the effect of the estimated $\hat{\nu}_i$. The multiplier bootstrap thus automatically captures the two-step nature of estimation without requiring analytical correction terms.

Under the conditions of Theorem 2, the conditional distribution of $\sqrt{n}(\theta^{(b)} - \hat{\theta})$ given the data converges weakly to $\mathcal{N}(0, V_\theta)$, ensuring first-order validity. This approach is computationally efficient, avoids second-step re-optimization, and is robust to the complex dependence structure encoded in Λ_2.

5. Application: Determinants of health care utilization

In this section, I investigate the determinants of health care utilization in Côte d'Ivoire using data from the 2018–2019 Harmonized Household Living Conditions Survey (EHCVM). The analysis focuses specifically on the type of health care professional consulted during an episode of illness. Health care utilization is influenced by a complex combination of economic, demographic, and contextual factors, and understanding these determinants is crucial for designing effective health policies and improving access to care. The outcome of interest is the type of health care professional first consulted during an illness episode, derived from the survey question: *Who did [name] first see during this episode of illness?* I categorize responses into three mutually exclusive and unordered groups:

(1) physician,
(2) other formal health care professional (such as nurses or midwives), and
(3) informal provider (such as traditional healers or marabouts).

This classification captures the most commonly reported patterns of provider choice while distinguishing between formal and informal sources of care. To analyze the determinants of provider choice, I estimate a multinomial logistic regression model, which allows modeling utilization across multiple unordered categories simultaneously. This framework accommodates the categorical nature of the outcome and avoids imposing an artificial ordering of provider types, which could lead to misleading inferences. The set of explanatory variables is designed to reflect the main drivers of health care demand documented in the literature. Specifically, economic resources are measured through the household wealth index, constructed via principal component analysis on asset ownership and housing characteristics (details provided in Appendix B). Human capital is proxied by the individual's highest attained education level. Demographic characteristics include sex and area of residence (urban versus rural). Together, these covariates provide a comprehensive view of how financial capacity, socioeconomic status, demographic attributes, and geographic location jointly influence the type of provider consulted. The sample is restricted to individuals aged 25–65 to focus on the working-age population and to reduce heterogeneity associated with pediatric and elderly care.

A key methodological challenge in this empirical context is sample selection. In the EHCVM, information on provider choice is only recorded for individuals who sought care

during an illness episode. For those who did not consult any provider, their hypothetical choice remains unobserved. If the decision to seek care is correlated with unobserved factors that also influence provider choice, such as underlying health status, cultural attitudes toward modern versus traditional medicine, or unmeasured household resources, then estimating the multinomial logit model using only the subsample of health care users will produce biased estimates. Formally, this corresponds to a sample selection problem where the observed provider distribution reflects the conditional distribution, rather than the unconditional distribution in the full population.

To address this issue, I exploit an instrumental variable available in the survey: the reported distance from the household to the location of the first consultation. Distance is a widely recognized determinant of health care access and strongly affects the probability of seeking any provider. Crucially, conditional on seeking care, distance is plausibly exogenous to the unobserved determinants of provider choice. For instance, while living closer to a facility increases the likelihood of obtaining care, it is unlikely to directly affect whether a person chooses a physician versus a traditional healer once they decide to seek care. This exclusion restriction allows us to isolate the selection process and correct for the truncation of the sample, thereby recovering consistent estimates of the effects of covariates on provider choice. The baseline outcome category in the multinomial logit is set tO informal provider, so that all coefficients are interpreted relative to this category.

The instrument also has the practical advantage of being directly measurable and easily interpretable. By leveraging variation in access induced by geographic proximity, we can identify the local dependence between selection into care and provider choice without relying on strong functional form assumptions. This approach is conceptually similar to the treatment of selection in classical sample selection models, but adapted to a multinomial outcome using the local representation developed in this paper. The estimated results are reported in Table 1. They provide insights into how wealth, education, sex, and urban residence shape the choice of provider, while accounting for the non-random selection into care. For instance, higher household wealth and education are associated with a greater probability of consulting a physician relative to an informal provider, and urban residents are more likely to choose formal providers overall. Importantly, the inclusion of the instrumental variable ensures that these estimates are robust to selection bias, allowing us to interpret the results

as reflecting true preferences for provider type, rather than being driven by differential access to care.

Table 1. Determinants of Health Care Provider Choice with Selection Correction

Parameter	**Estimate**	**95% CI Lower**	**95% CI Upper**
Provider Choice Equation (relative to informal provider)			
Physician (Category 1)			
Wealth	1.2828	1.2819	1.2837
Education	2.6222	2.6212	2.6231
Sex (male=1)	2.0061	2.0055	2.0067
Urban area	1.9906	1.9900	1.9911
Formal Provider (Category 2)			
Wealth	1.1957	1.1948	1.1965
Education	2.6173	2.6164	2.6182
Sex (male=1)	1.9036	1.9030	1.9042
Urban area	2.4415	2.4410	2.4419
Sorting parameters			
Physician (Category 1)			
Wealth	-0.0065	-0.0065	-0.0065
Education	1.0421	1.0421	1.0421
Sex (male=1)	3.0409	3.0409	3.0409
Urban area	5.7665	5.7665	5.7665
Formal Provider (Category 2)			
Wealth	0.5264	0.5257	0.5270
Education	-0.0253	-0.0258	-0.0248
Sex (male=1)	2.2109	2.2101	2.2117
Urban area	1.2356	1.2352	1.2360

Notes: The table reports estimated coefficients and 95% confidence intervals from a multinomial logit model of provider choice with correction for selection into care-seeking. Category (i) is physician, (ii) other formal provider, and (iii) informal provider (reference). Covariates include wealth (log total annual consumption), education, sex, and urban area indicator.

The results highlight strong and systematic associations between socioeconomic and demographic characteristics and the type of health care provider consulted in Côte d'Ivoire.

Higher wealth and education both substantially increase the likelihood of consulting a physician or another formal provider rather than an informal one. The magnitude of the coefficients is particularly high for education 2.6, suggesting that education is a key channel shaping health care choices, possibly through greater awareness of medical quality or trust in modern health care. Gender differences are also pronounced. Males are more likely to consult formal providers, with coefficients around 2.0 in the provider equations. This pattern could reflect gender disparities in autonomy or access to financial resources for health care. Geographical location exerts a strong effect. Living in an urban area is associated with a much higher probability of consulting formal care, consistent with better availability of facilities and professionals. This indicates that proximity to care facilities strongly predicts whether individuals seek care at all.

Turning to the selection equations, the results reveal clear evidence of non-random sorting into health care utilization. Individuals with higher education and those residing in urban areas are significantly more likely to seek any form of care, as reflected by the large positive coefficients on these variables. In contrast, the wealth effect appears relatively modest and even slightly negative in one specification, suggesting that financial resources play a smaller role in the initial decision to seek care once other factors are controlled for. These findings underscore that the observed sample of health care users is not representative of the general population: more educated and urban individuals are overrepresented among those who seek treatment. Accounting for this selection process is therefore essential, as it ensures that estimated differences in provider choice reflect true behavioral responses rather than compositional differences driven by who sorts into the care-seeking sample.

Overall, the analysis provides robust evidence that both financial and spatial access, along with education and gender, are key determinants of health care utilization patterns in Côte d'Ivoire. By correcting for selection into care, the estimates reflect more accurately the structural drivers of provider choice rather than compositional differences in who seeks care.

6. Conclusion

This paper develops a flexible framework for addressing sample selection in settings with categorical outcomes. Building on the ideas of the Local Gaussian Representation, I introduce a local representationthat applies to continuous, ordinal, and nominal outcomes. This representation decomposes joint probabilities into marginal probabilities and category-specific local association parameters, allowing us to capture how selection affects each outcome without imposing ordering or strong parametric assumptions. Leveraging this tool, I establish nonparametric point identification of latent categorical distributions under exclusion restrictions. I also propose a semiparametric multinomial logistic regression model with sample selection, accompanied by a practical two-step estimation procedure and asymptotic theory. Our approach broadens the empirical toolkit for analyzing selection bias in categorical and discrete outcomes. It provides a flexible and interpretable method that can be applied in a variety of contexts, from labor market studies to health and education research, where unordered categorical outcomes are common.

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7. Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proofs of the results in the main text

A.1. **Proof of Lemma 1.** :

Y, S are any random variables with support that respectively are the measurable sets $(\mathcal{Y}, \sigma_Y), (\mathcal{S}, \sigma_S)$. For all pairs of sets $A \times B \in \sigma_Y \times \sigma_S$ with positive probabilities. Let $X = \mathbf{1}_{\{Y \in A\}}$ and $Z = \mathbf{1}_{\{S \in B\}}$. Write

$$u = \mathbb{E}[X] = \mathbb{P}(Y \in A) \in (0,1), \qquad v = \mathbb{E}[Z] = \mathbb{P}(S \in B) \in (0,1).$$

Compute the covariance

$$\mathrm{Cov}(X, Z) = \mathbb{E}[XZ] - \mathbb{E}[X]\mathbb{E}[Z] = \mathbb{P}(Y \in A,\, S \in B) - uv.$$

Also note the variances

$$\mathrm{Var}(X) = u(1-u), \qquad \mathrm{Var}(Z) = v(1-v),$$

which are strictly positive under the assumption $u, v \in (0,1)$.

Define

$$\omega_{Y,S}(A,B) := \mathrm{Corr}(X,Z) = \frac{\mathrm{Cov}(X,Z)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Z)}} = \frac{\mathbb{P}(Y \in A,\, S \in B) - pq}{\sqrt{u(1-u)}\sqrt{v(1-v)}}.$$

By the Cauchy–Schwarz inequality (or basic properties of correlation), $\omega_{Y,S}(A,B) \in [-1,1]$. Rearranging the display above yields the desired representation:

$$\mathbb{P}(Y \in A,\, S \in B) = uv + \omega_{Y,S}(A,B)\,\sqrt{u(1-u)}\sqrt{v(1-v)} = \Lambda_2\big(u, v, \omega_{Y,S}(A,B)\big).$$

Uniqueness follows immediately: since $\sqrt{u(1-u)}\sqrt{v(1-v)} > 0$, the value of $\omega_{Y,S}(A,B)$ is determined uniquely by the observed joint probability $\mathbb{P}(Y \in A,\, S \in B)$. Finally, the sign interpretation is immediate from the definition of correlation: $\omega > 0$ iff $\mathrm{Cov}(X,Z) > 0$ (local

positive dependence), $\omega < 0$ iff $\mathrm{Cov}(X, Z) < 0$ (local negative dependence), and $\omega = 0$ iff $\mathrm{Cov}(X, Z) = 0$ (local independence).

A.2. **Proof of Theorem 1.** Define the parameter vector

$$\theta := (\mu_1, \ldots, \mu_{q-1}, \omega_1, \ldots, \omega_{q-1}) \in \Theta \subset \mathbb{R}^{2(q-1)},$$

where the domain Θ is the interior of the admissible set: $\mu_k \in \mathbb{R}$, and $\omega_k \in (-1, 1)$ (interiority). Define the observable vector of joint probabilities:

$$\mathbf{p} := \big(p_{k,z}\big)_{k=1,\ldots,q-1;\, z=0,1} \in \mathbb{R}^{2(q-1)}, \quad \text{where } p_{k,z} := \mathbb{P}(Y = c_k, S = 1 \mid Z = z).$$

The model implies the structural mapping $\mathbf{F} : \Theta \to \mathbb{R}^{2(q-1)}$ given by

$$F_{k,z}(\theta) := \Lambda_2\big(\pi_k(\mu),\, \nu_z,\, \omega_k\big) = \pi_k(\mu)\, \nu_z + \omega_k \sqrt{\pi_k(\mu)(1 - \pi_k(\mu))\, \nu_z(1 - \nu_z)}.$$

We aim to show that $\mathbf{F}$ is globally injective on Θ. By the Gale–Nikaido Theorem (Gale and Nikaido (1965)), it suffices to verify that: the Jacobian matrix $J(\theta) := D\mathbf{F}(\theta)$ is a P-matrix for all $\theta \in \Theta$ (i.e., all its principal minors are strictly positive).

Because the system decouples across categories—$F_{k,z}$ depends only on μ_k and ω_k, not on $\mu_{k'}$ or $\omega_{k'}$ for $k' \neq k$—the Jacobian is block-diagonal with $q - 1$ blocks of size 2×2, one per category k. Thus, it suffices to show that each 2×2 block is a P-matrix (i.e., has positive diagonal entries and positive determinant).

Compute the Jacobian block for fixed k. Let us fix k and suppress the subscript for clarity. Define: $\pi := \pi_k(\mu_k) = \frac{e^{\mu_k}}{1+\sum_{j=1}^{q-1} e^{\mu_j}}$. However, note that even though π_k depends on all μ_j in the multinomial logit, under standard identification of multinomial logit models, we treat the baseline as fixed and consider π as a function of the full μ. We reparameterize the model in terms of $\pi = (\pi_1, \ldots, \pi_{q-1}) \in \mathcal{P}$, where $\mathcal{P} = \{\pi \in (0, 1)^{q-1} : \sum_{k=1}^{q-1} \pi_k < 1\}$, and treat π (not μ) as the structural parameter. This is permissible because: the mapping $\mu \mapsto \pi$ is a diffeomorphism between $\mathbb{R}^{q-1}$ and $\mathcal{P}$, point identification of π implies point identification of μ.

So for the purpose of proving injectivity of $\mathbf{F}$, we work with the reparameterized mapping

$$\tilde{\mathbf{F}} : \tilde{\Theta} \to \mathbb{R}^{2(q-1)}, \quad \tilde{\theta} = (\pi_1, \ldots, \pi_{q-1}, \omega_1, \ldots, \omega_{q-1}),$$

with

$$\tilde{F}_{k,z}(\tilde{\theta}) = \pi_k \nu_z + \omega_k \sqrt{\pi_k(1 - \pi_k)\nu_z(1 - \nu_z)}.$$

Now, $\tilde{F}_{k,z}$ depends only on π_k and ω_k, then the Jacobian is block-diagonal. For fixed k, define $g_k(\pi_k, \omega_k) = \left(\tilde{F}_{k,0}, \tilde{F}_{k,1}\right)^\top$. The 2×2 Jacobian block is:

$$J_k = \begin{bmatrix} \frac{\partial \tilde{F}_{k,0}}{\partial \pi_k} & \frac{\partial \tilde{F}_{k,0}}{\partial \omega_k} \\ \frac{\partial \tilde{F}_{k,1}}{\partial \pi_k} & \frac{\partial \tilde{F}_{k,1}}{\partial \omega_k} \end{bmatrix} = \begin{bmatrix} \nu_0 + \omega_k \cdot A_0 & B_0 \\ \nu_1 + \omega_k \cdot A_1 & B_1 \end{bmatrix},$$

where: $B_z = \sqrt{\pi_k(1-\pi_k)\nu_z(1-\nu_z)} > 0$ (since $\pi_k \in (0,1)$, $\nu_z \in (0,1)$), $A_z = \frac{\partial}{\partial \pi_k}\sqrt{\pi_k(1-\pi_k)} \cdot \sqrt{\nu_z(1-\nu_z)} = \frac{1-2\pi_k}{2\sqrt{\pi_k(1-\pi_k)}} \cdot \sqrt{\nu_z(1-\nu_z)}$.

First-order minors: $\frac{\partial \tilde{F}_{k,z}}{\partial \omega_k} = B_z > 0$

Second-order minor (determinant):

$$\det(J_k) = (\nu_0 + \omega_k A_0)B_1 - (\nu_1 + \omega_k A_1)B_0.$$

Simplify:

$$\det(J_k) = \sqrt{\pi_k(1-\pi_k)} \cdot \left(\nu_0\sqrt{\nu_1(1-\nu_1)} - \nu_1\sqrt{\nu_0(1-\nu_0)}\right).$$

As argued earlier, if $\nu_0 \neq \nu_1$ (instrument relevance), then $\nu_0 t_1 - \nu_1 t_0 \neq 0$. Without loss of generality, assume $\nu_1 > \nu_0$; then one can verify that $\det(J_k) > 0$. (For example, define $h(x) = x/\sqrt{x(1-x)} = \sqrt{x/(1-x)}$, which is increasing; then $\nu_1 > \nu_0 \Rightarrow h(\nu_1) > h(\nu_0) \Rightarrow \nu_1 t_0 > \nu_0 t_1$.)

However, to guarantee global positivity, we can use the following fact: the mapping $(\pi_k, \omega_k) \mapsto (p_{k,0}, p_{k,1})$ is the composition of a linear invertible map (as shown in the original proof) with a smooth reparameterization. Since we already know the mapping is bijective (from the explicit solution), and continuously differentiable, global univalence holds—and Gale–Nikaido provides sufficient but not necessary conditions.

By the Gale–Nikaido theorem, since the Jacobian of $\tilde{\mathbf{F}}$ is a P-matrix at every point in the interior parameter space (as each block has positive determinant and positive diagonal entries under interiority and instrument relevance), the mapping $\tilde{\mathbf{F}}$ is globally univalent. Hence, the system (1) has at most one solution. Existence follows from the data-generating process satisfying the model. Therefore, the parameters (π, ω) and hence (μ, ω) are point-identified.

A.3. **Proof of theorem 2.** We proceed in two main steps: consistency and asymptotic normality. The estimator $\hat{\theta}$ solves a second-stage M-estimation problem that depends on a first-stage estimator $\hat{\delta}$ of δ_0.

Let the population second-stage criterion be

$$Q_0(\theta,\delta) := \mathbb{E}\big[\ell_i(\theta,\delta)\big],$$

where

$$\ell_i(\theta,\delta) = \sum_{k=1}^{q} S_i \mathbf{1}\{Y_i = c_k\} \log \Lambda_2\big(\pi_k(X_i;\beta),\, \Lambda(W_i'\delta),\, \tanh(X_i'\gamma_k)\big).$$

By Assumption (i), the data are generated from the model at (θ_0,δ_0), so (θ_0,δ_0) uniquely maximizes $Q_0(\theta,\delta)$ under the identification conditions implied by Theorem 1 and Assumption 2. In particular, Lemma 1 ensures that the representation via Λ_2 is exact and invertible on the interior of the parameter space.

By Assumption (ii), the parameter space for θ is compact and θ_0 is interior. Assumptions (iv)–(vi) ensure that $\ell_i(\theta,\delta)$ is continuous in (θ,δ) and dominated by an integrable function. Hence, by the Uniform Law of Large Numbers (ULLN),

$$\sup_{\theta,\delta} \Big|\frac{1}{n}\sum_{i=1}^{n} \ell_i(\theta,\delta) - Q_0(\theta,\delta)\Big| \xrightarrow{p} 0.$$

From the first-step logit estimation, standard results imply that under Assumptions (i)–(iii) and (vi),

$$\hat{\delta} \xrightarrow{p} \delta_0.$$

Now consider the second-stage estimator

$$\hat{\theta} = \arg\max_{\theta} \frac{1}{n}\sum_{i=1}^{n} \ell_i(\theta,\hat{\delta}).$$

Since $\hat{\delta} \xrightarrow{p} \delta_0$ and $Q_0(\theta,\delta)$ is continuous and uniquely maximized at θ_0 when $\delta = \delta_0$, a standard stochastic equicontinuity argument (e.g., Theorem 2.1 in Newey and McFadden (1994)) yields

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

We now derive the asymptotic distribution of $\sqrt{n}(\hat{\theta}-\theta_0)$. By a mean-value expansion of the first-order condition for $\hat{\theta}$, we have

$$0 = \frac{1}{n}\sum_{i=1}^{n} \nabla_\theta \ell_i(\hat{\theta},\hat{\delta}) = \frac{1}{n}\sum_{i=1}^{n} \nabla_\theta \ell_i(\theta_0,\delta_0) + \underbrace{\left[\frac{1}{n}\sum_{i=1}^{n} \nabla^2_{\theta\theta}\ell_i(\bar{\theta},\bar{\delta})\right]}_{-\hat{H}_2}(\hat{\theta}-\theta_0) + \underbrace{\left[\frac{1}{n}\sum_{i=1}^{n} \nabla^2_{\theta\delta}\ell_i(\tilde{\theta},\tilde{\delta})\right]}_{-\hat{G}}(\hat{\delta}-\delta_0),$$

for some $\bar{\theta} \to_p \theta_0$, $\bar{\delta} \to_p \delta_0$, $\tilde{\theta} \to_p \theta_0$, $\tilde{\delta} \to_p \delta_0$ by the mean-value theorem.

Rearranging,

$$\hat{H}_2(\hat{\theta} - \theta_0) = \frac{1}{n}\sum_{i=1}^{n} \psi_{2,i} - \hat{G}(\hat{\delta} - \delta_0) + o_p(n^{-1/2}),$$

where $\psi_{2,i} = \nabla_\theta \ell_i(\theta_0, \delta_0)$.

By Assumptions (iv)–(vi) and the ULLN,

$$\hat{H}_2 \xrightarrow{p} H_2 = -\mathbb{E}[\nabla^2_{\theta\theta}\ell_i(\theta_0, \delta_0)] \succ 0, \quad \hat{G} \xrightarrow{p} G = -\mathbb{E}[\nabla^2_{\theta\delta}\ell_i(\theta_0, \delta_0)].$$

From the first-step logit estimation, the asymptotic linear representation is

$$\sqrt{n}(\hat{\delta} - \delta_0) = I_1^{-1} \cdot \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_{1,i} + o_p(1), \quad \psi_{1,i} = (S_i - \Lambda(W_i'\delta_0))W_i,$$

and $I_1 = \mathbb{E}[\Lambda(W'\delta_0)(1 - \Lambda(W'\delta_0))WW'] \succ 0$ by Assumption (iii).

Substituting into the expansion for $\hat{\theta}$, we obtain

$$\sqrt{n}(\hat{\theta} - \theta_0) = H_2^{-1}\left[\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_{2,i} - G \cdot I_1^{-1} \cdot \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_{1,i}\right] + o_p(1).$$

Define the combined influence function

$$\phi_i = \psi_{2,i} - GI_1^{-1}\psi_{1,i}.$$

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) = H_2^{-1} \cdot \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \phi_i + o_p(1).$$

By the Lindeberg–Lévy Central Limit Theorem, and under the moment conditions in Assumption (vi),

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \phi_i \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\phi_i\phi_i^\top]).$$

Now compute the asymptotic variance:

$$\mathbb{E}[\phi_i\phi_i^\top] = \mathbb{E}[\psi_{2,i}\psi_{2,i}^\top] - GI_1^{-1}\mathbb{E}[\psi_{1,i}\psi_{2,i}^\top] - \mathbb{E}[\psi_{2,i}\psi_{1,i}^\top]I_1^{-1}G^\top + GI_1^{-1}\mathbb{E}[\psi_{1,i}\psi_{1,i}^\top]I_1^{-1}G^\top.$$

But note that under correct specification of the first-step model,

$$\mathbb{E}[\psi_{1,i}\psi_{2,i}^\top] = \mathbb{E}\big[(S_i - \nu_i)W_i \cdot \nabla_\theta \ell_i(\theta_0, \delta_0)\big] = 0,$$

because $\psi_{2,i}$ depends on S_i only through the event $\{Y_i = c_k, S_i = 1\}$, and conditional on W_i, S_i is independent of the structural errors in the outcome under the model (by construction

via the Λ_2 representation). More formally, in the correctly specified model, the score $\psi_{2,i}$ is mean-independent of $\psi_{1,i}$ due to the orthogonal parameterization in the local representation.

Thus, the cross terms vanish, and since $\mathbb{E}[\psi_{1,i}\psi_{1,i}^\top] = I_1$, we obtain

$$\mathbb{E}[\phi_i\phi_i^\top] = \Omega + GI_1^{-1}G^\top, \quad \text{where } \Omega = \mathbb{E}[\psi_{2,i}\psi_{2,i}^\top].$$

Therefore,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\big(0,\, H_2^{-1}(\Omega + GI_1^{-1}G^\top)H_2^{-1}\big) = \mathcal{N}(0, V_\theta).$$

Appendix B. Wealth index

We also have information on the sets of assets possessed by the household, and we use these variables to construct a simple asset index. This is done in a straightforward manner for physical assets, by a simple weighted average of the binary responses corresponding to whether or not a given list of items are owned by a household, using weights from a principal components analysis (PCA).

$$asset = \sum_{i=1}^{n} \omega_i b_i$$

We use 18 binary variables in this application: wall in good material, roof in good material, floor in good material, access to drinking water in the dry season and in the rainy season, access to electricity, use of the electricity network, clean restrooms, good waste management, good waste water management, owns a TV, has an iron, has a refrigerator, has a stove, has a computer, has cable, owns a car.

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